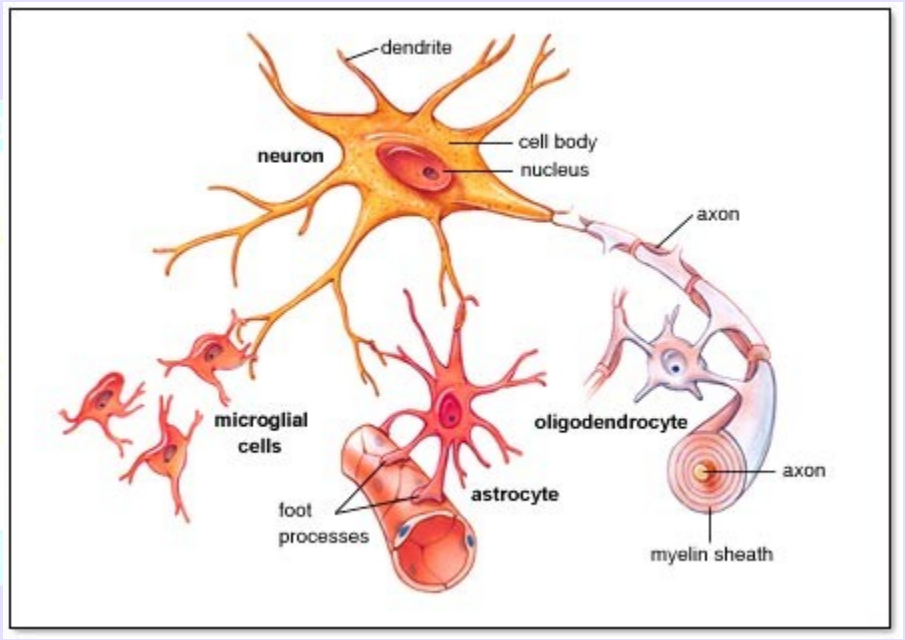
HISTOLOGY REVISIT: NEURONS AND NEUROGLIA

LEARNING OBJECTIVES

- At the end of class the student should be able,
- To recall the previous knowledge of the nerve cells
- Discuss the types of neurons
- Describe the parts of neurons
- Recall the supporting cell of nervous system, their type and functions

Nervous tissue consists of

1. Nerve cells
2. Neuroglial cells

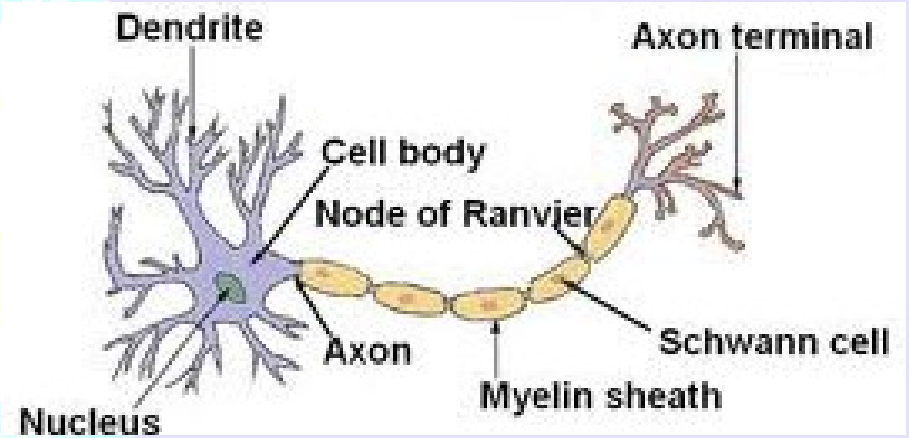


NERVE CELL

- Nerve cell is also called neuron and is characterized by its conductivity and irritability

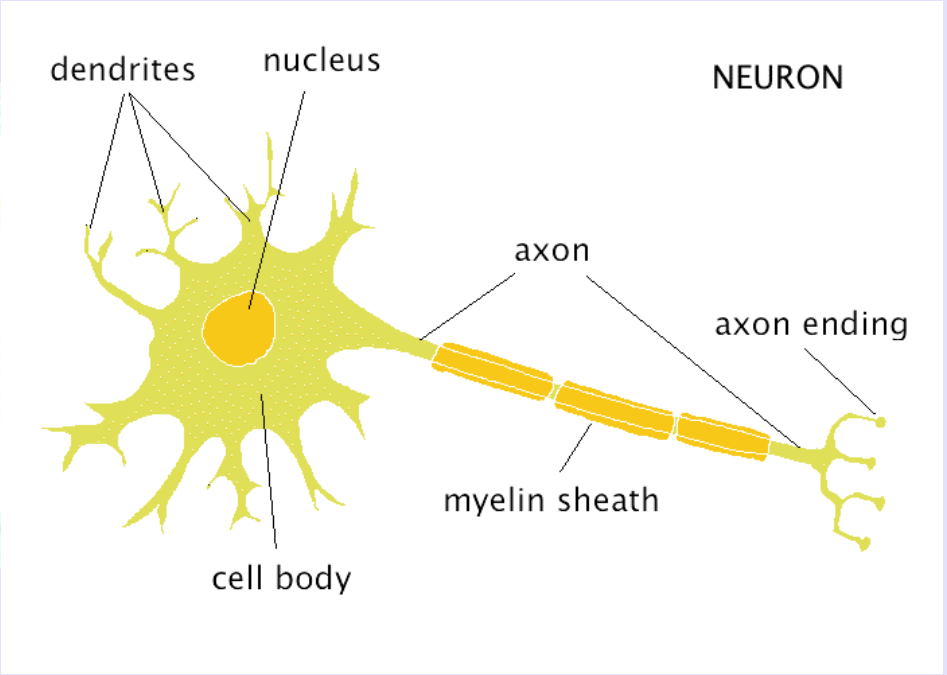
- **Conductivity i**s the ability of the nerve cell to transmit the impulses along its processes

- **Excitability i**s the ability of the nerve cell to initiate impulse in response to physical or chemical agents

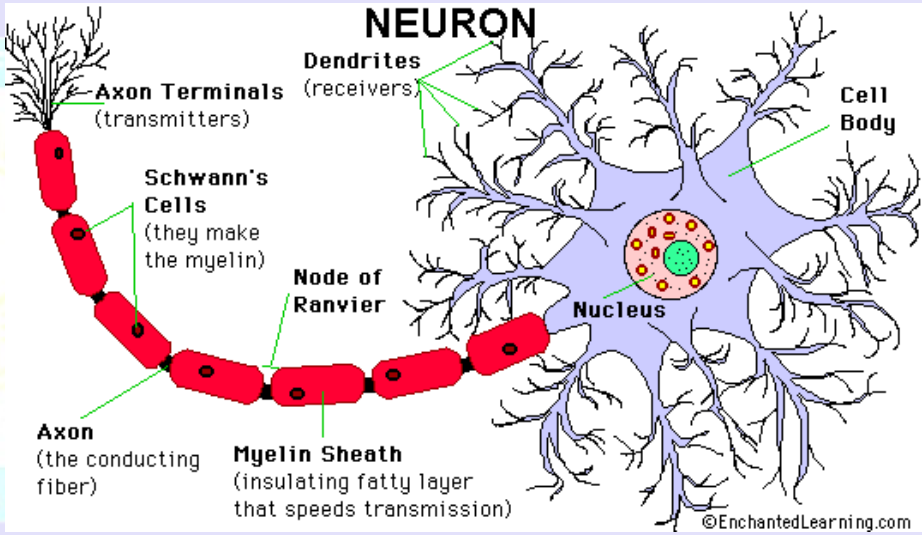


NEURON

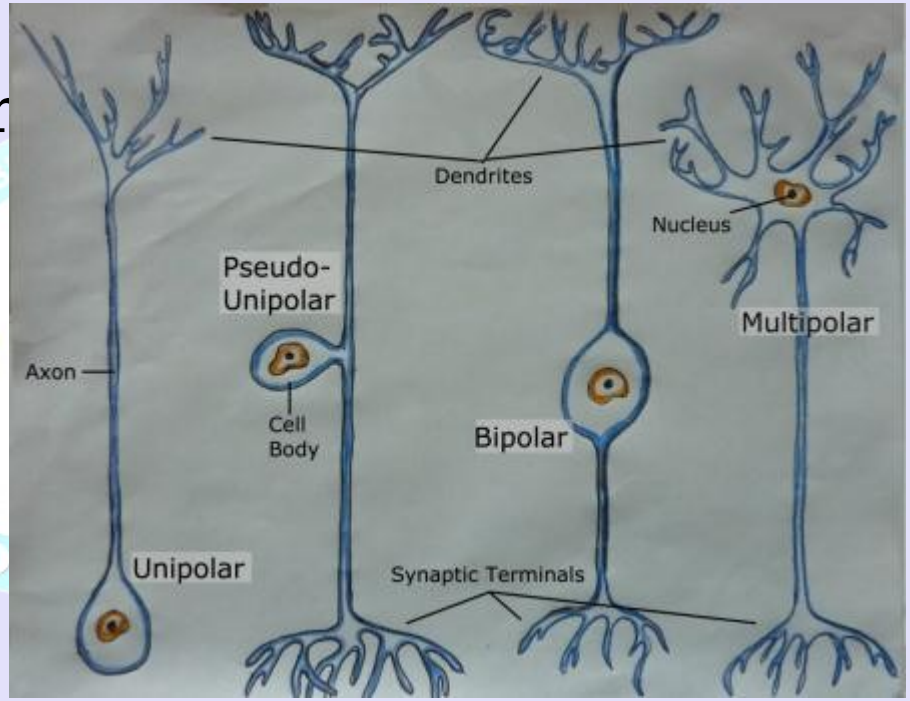
- Structural and functional unit of the nervous system is neuron
- Neuron includes cell body and all of its processes
- Neuron is like other cells of body but with certain exceptions



- Cell body is also called **soma**
- Processes of neuron are called **neurites** and consists of
- **Axon** is the process which carry nerve impulses away from the cell body.
- **Dendrites** receives impulses and convey them to soma

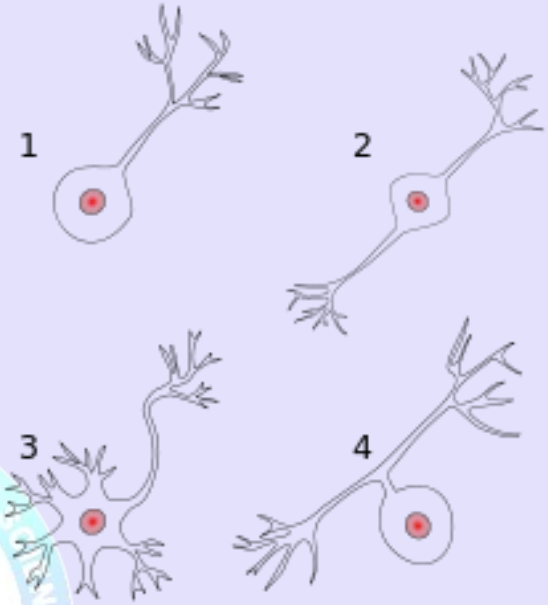


- **According to morphological classification neurons are classified into four types**

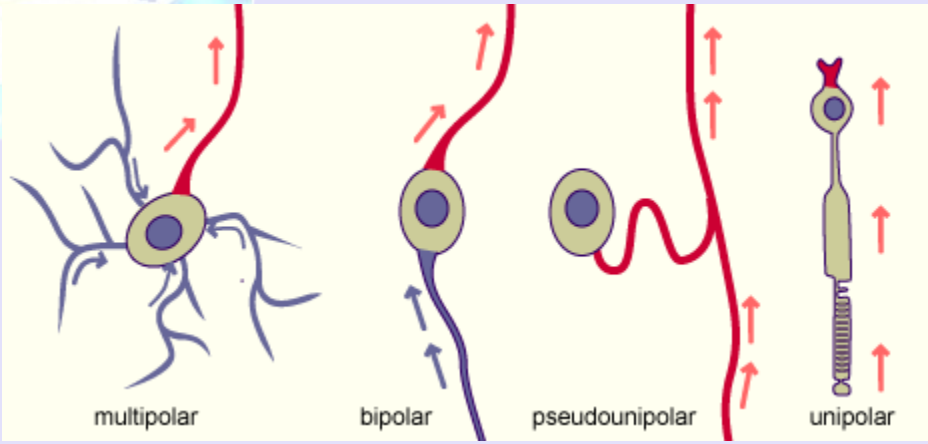


- A. Unipolar neuron
- B. Bipolar neuron
- C. Pseudounipolar neuron
- D. Multipolar neuron

- **Unipolar neuron**:
- Have one process, functionally axon e.g. mesencephelic nucleus of trigeminal nerve

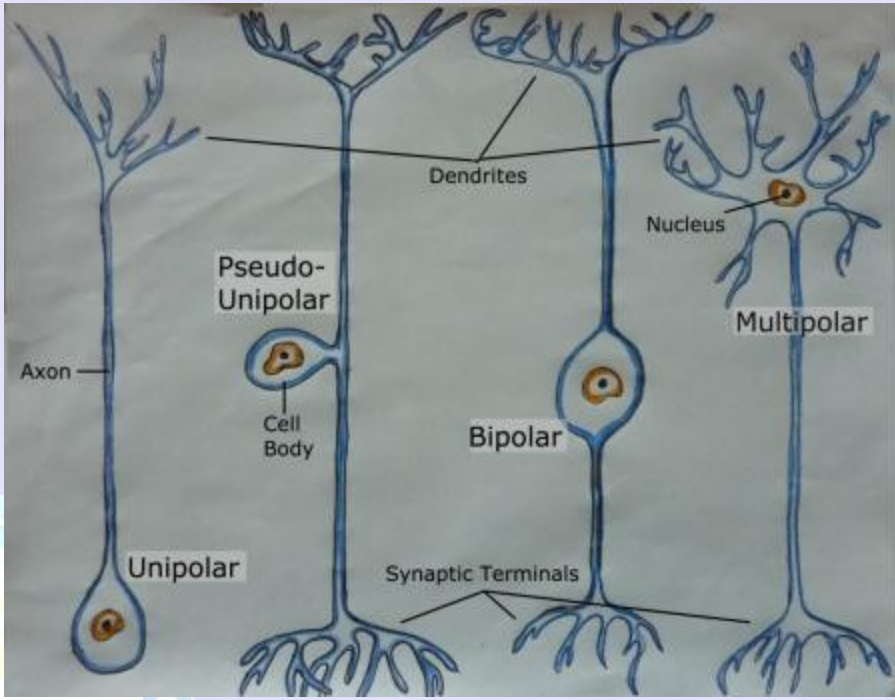


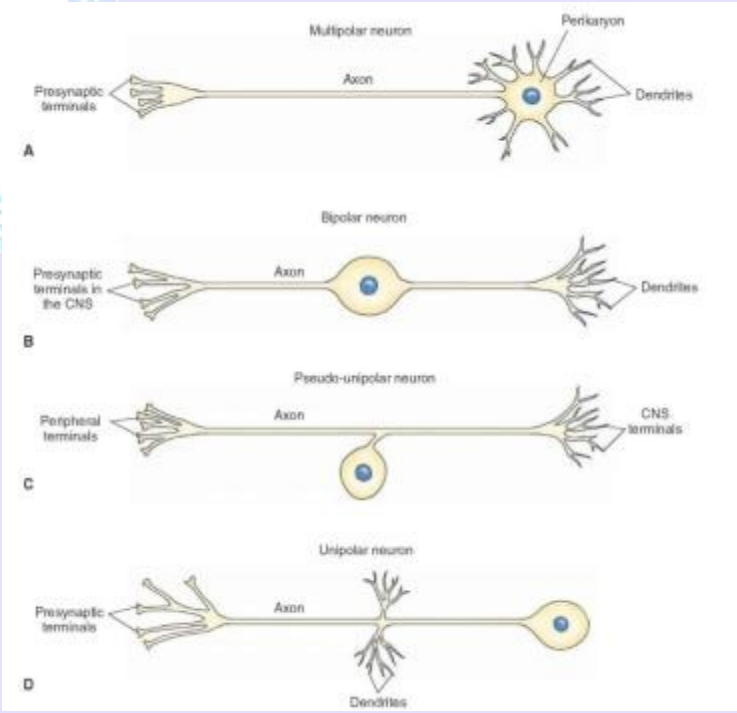
- **Bipolar neuron**
- Single axon and single dendrite
- Both arise from opposite ends of spindle shaped body e.g. bipolar neurons which are present in olfactory, optic and vestibulocochlear nerve



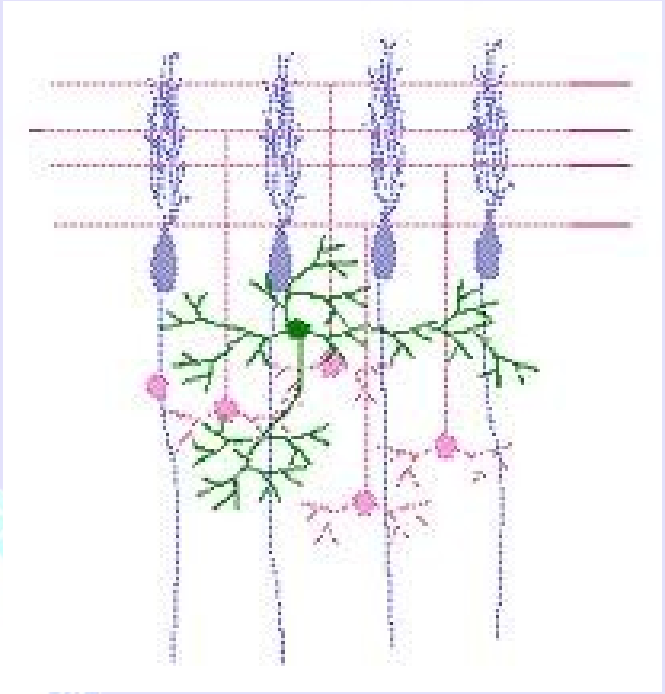
- **Pseudounipolar neuron**
- Posses a single process which then divides into dendrite and axon e.g. dorsal root ganglia of spinal nerves and sensory ganglia of cranial nerves

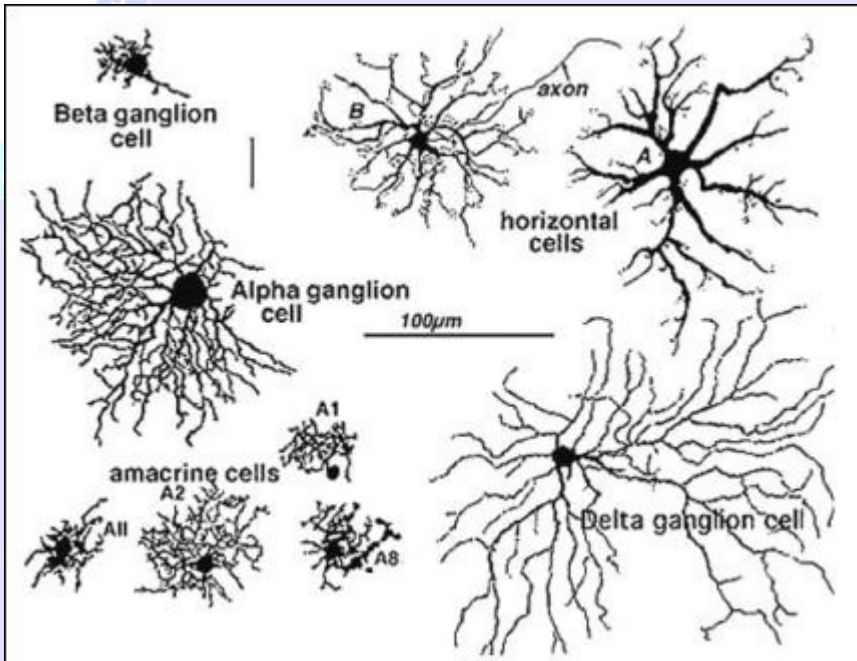
- **Multipolar neuron**
- Many dendrites arise from the cell body or soma of the neuron.
- A long single axon arise from the body
- Mostly neurons belong to multipolar variety e.g. pyramidal cells of cerebral cortex, anterior horn cells of spinal cord.





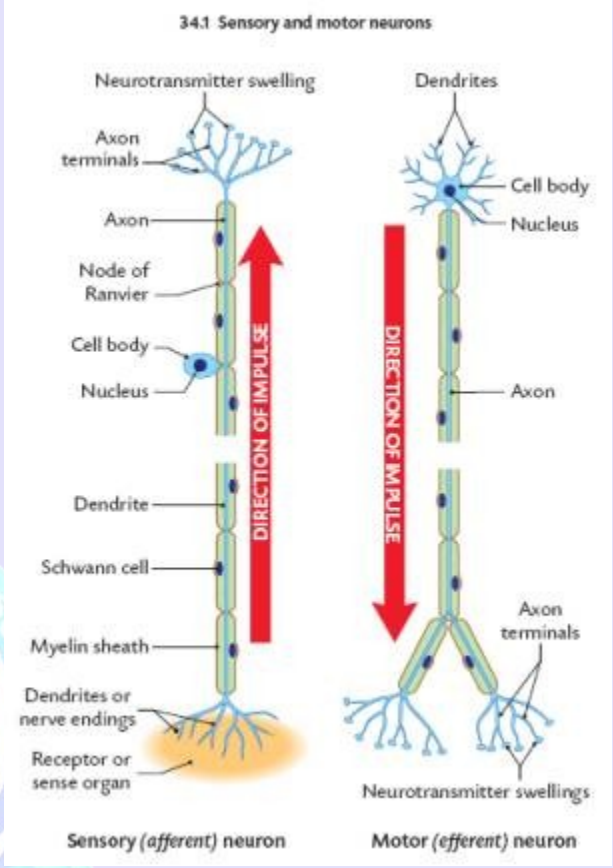
- **Classification of neurons according to the length of their axons**
- **A. Golgi type 1 neuron**
- Very long axon which leaves central nervous system and passes to other regions in the CNS or as nerve fiber to PNS, for example, pyramidal cells of cerebral cortex and anterior horn cells of spinal cord.
- **B. Golgi type 2 neurons**
- It has short axon which does not leave that part of gray matter in which the cell body of the neuron lies.
- It has many branched dendrites example, neurons present in cerebral and cerebellar cortices, these are mostly interneurons and are inhibitory in function.

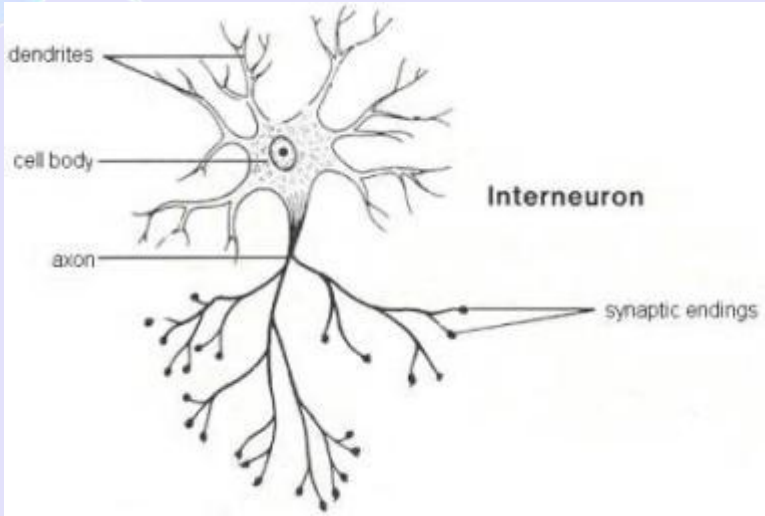




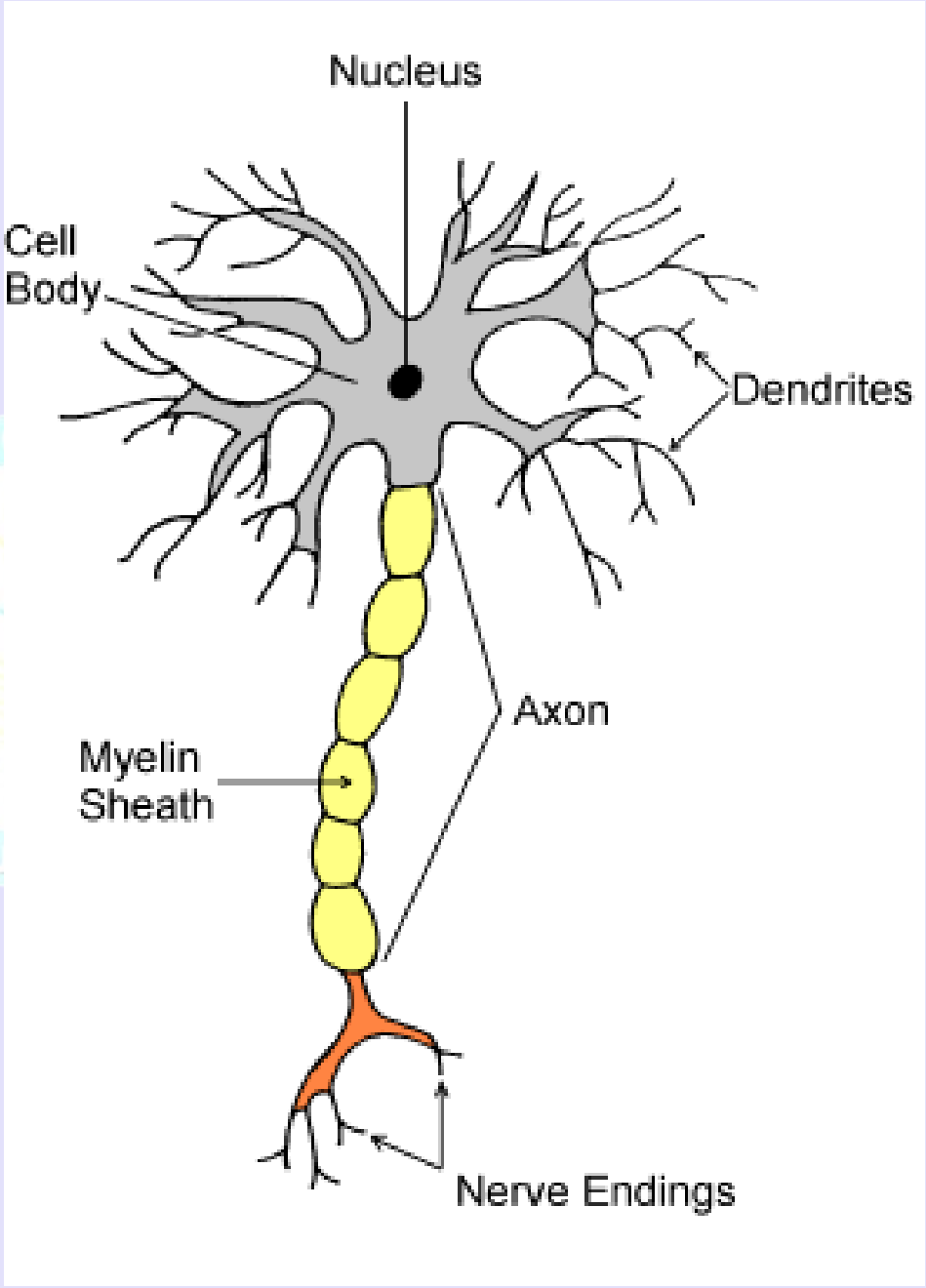
Functional classification of neurons.

- **A. Sensory neurons**
- Receive sensory information from environment and from with in body and pass them to CNS.
- **B. Motor neurons_**
- Are responsible for the control of the effecter organs such as muscles and glands
- **C. Interneurons_**
- These control other neurons to establish





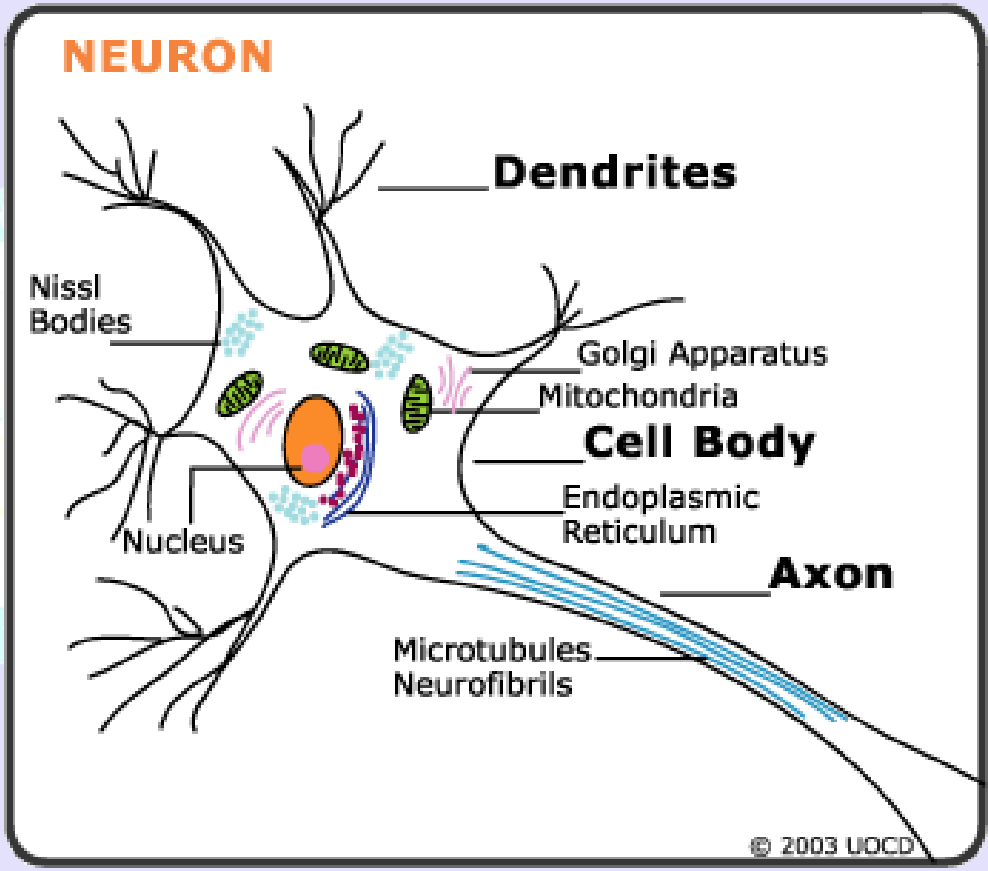
- **Structure of the neuron consists of**
- A: Cell body
- B: Processes



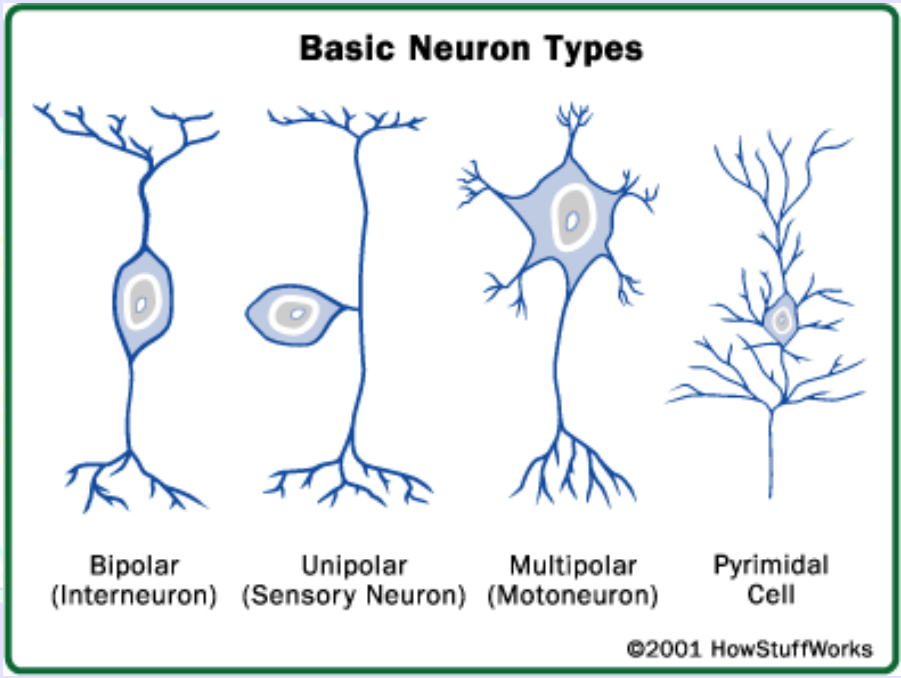
- **Cell body** of neuron is also called perikarya or soma which contains

1. Nucleus
2. Organelles
3. Cytoplasm

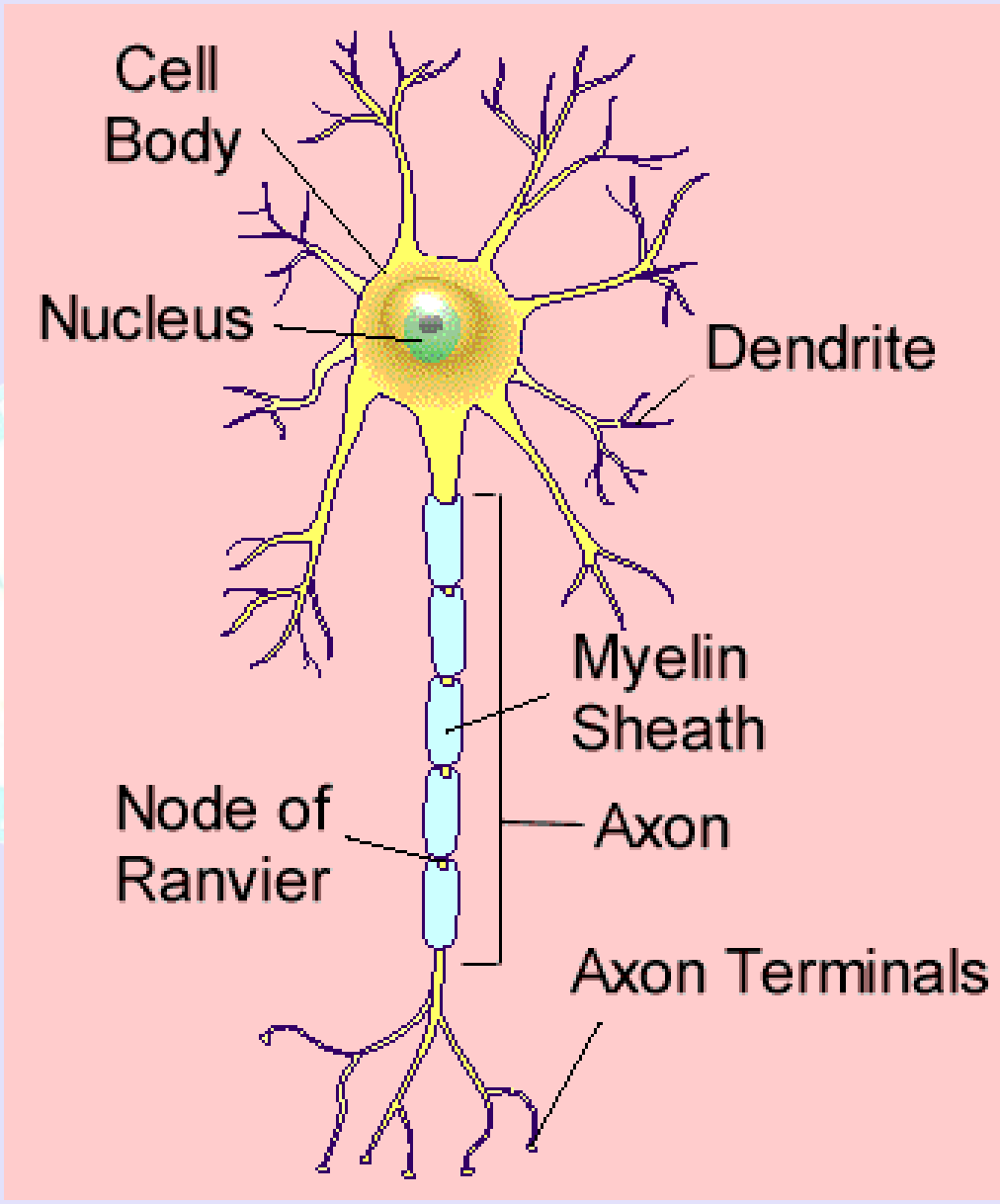
- **Cell body** varies in size from 4 micrometer as in granule cells of cerebral cortex to I35 micrometer in anterior horn cells of



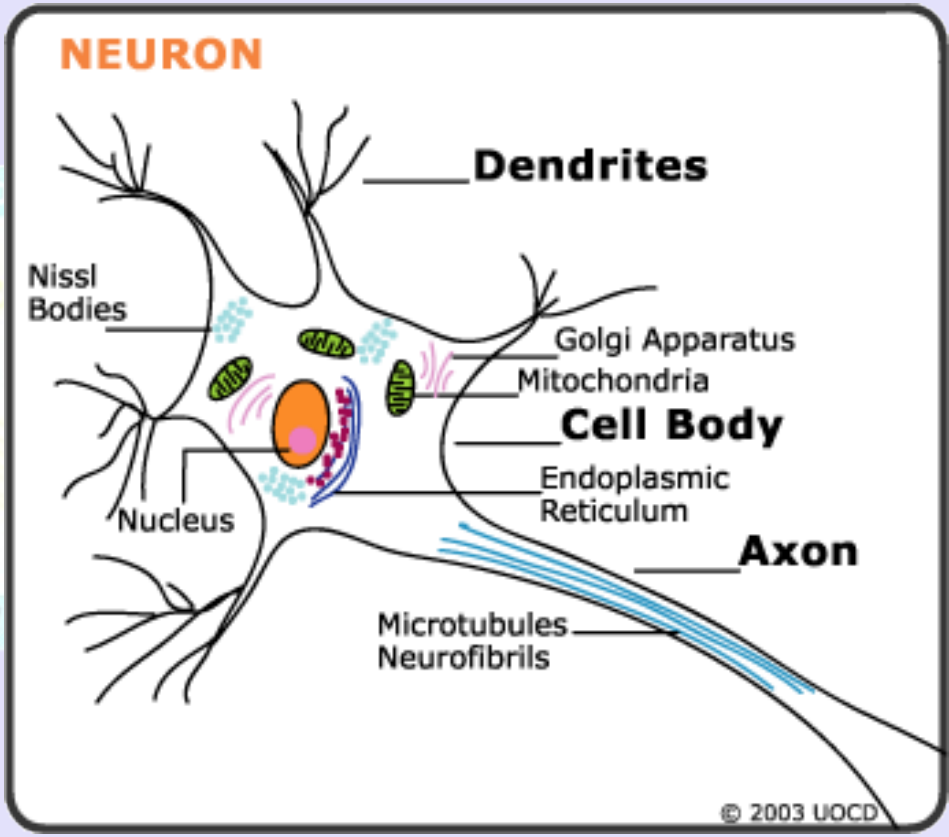
- Shape of body may be globular in pseudounipolar cells
- Spindle shaped in bipolar neurons
- Vary from pyramidal to globular in multipolar cells



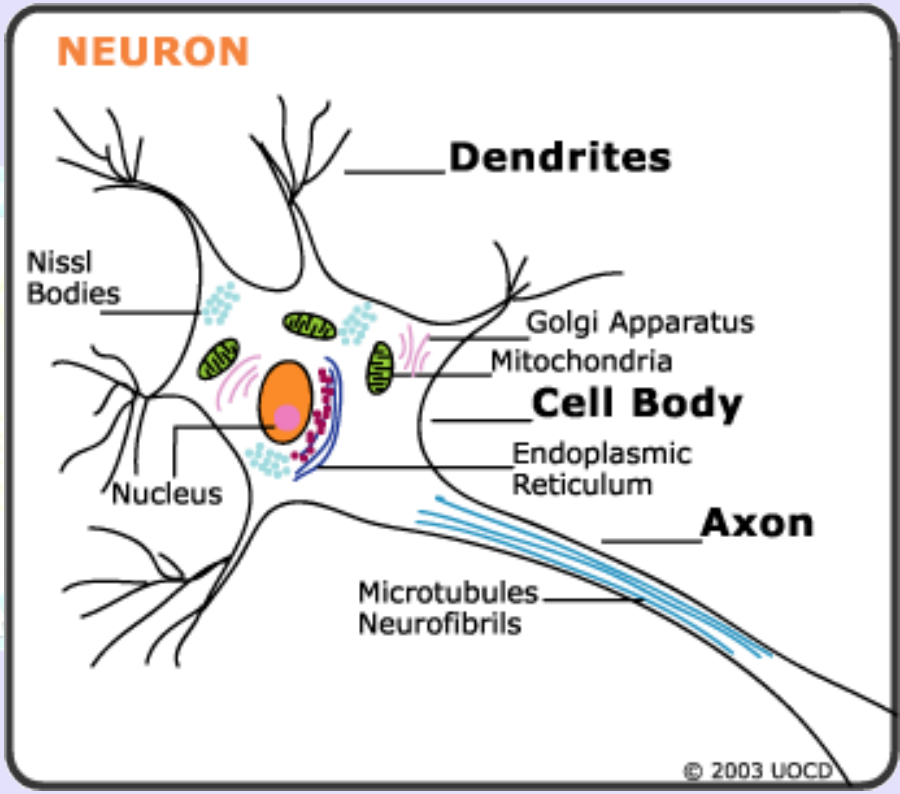
- **Nucleus**
- Single, large, pale staining (vesicular) indicates that chromatin is finely dispersed due to intense activity
- Spherical in shape and central in position
- Prominent nucleolus (owl's eyes)
- Barr body may be seen (rounded chromatin clump in females) close to nucleus



- **Cytoplasm:** Contains: organelles inclusions and elements of cytoskeleton
- **Mitochondria** are scattered through out the body
- **Golgi apparatus** is well developed and is around the nucleus



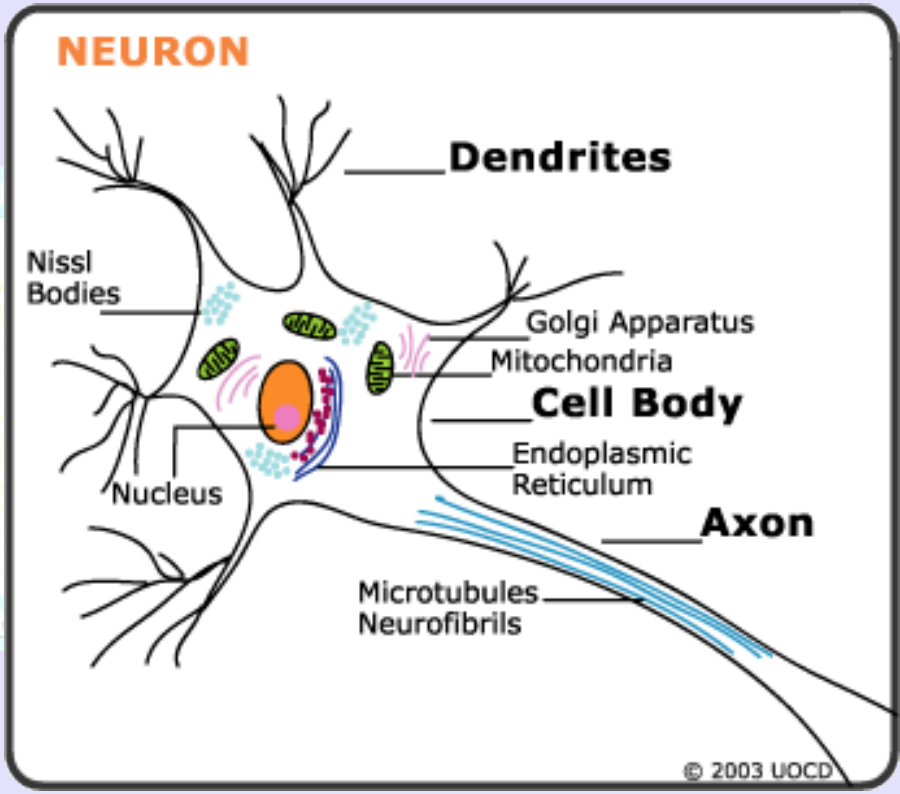
- **Rough endoplasmic reticulum** is highly developed and is scattered in body as aggregation of cisternae with polysomes.
- This arrangement suggestive of synthesis activity of neuron



- **Nissal substance**
- In stained sections rough endoplasmic reticulum and free ribosomes appear as patches of basophilic material called nissal substance or nissal granules

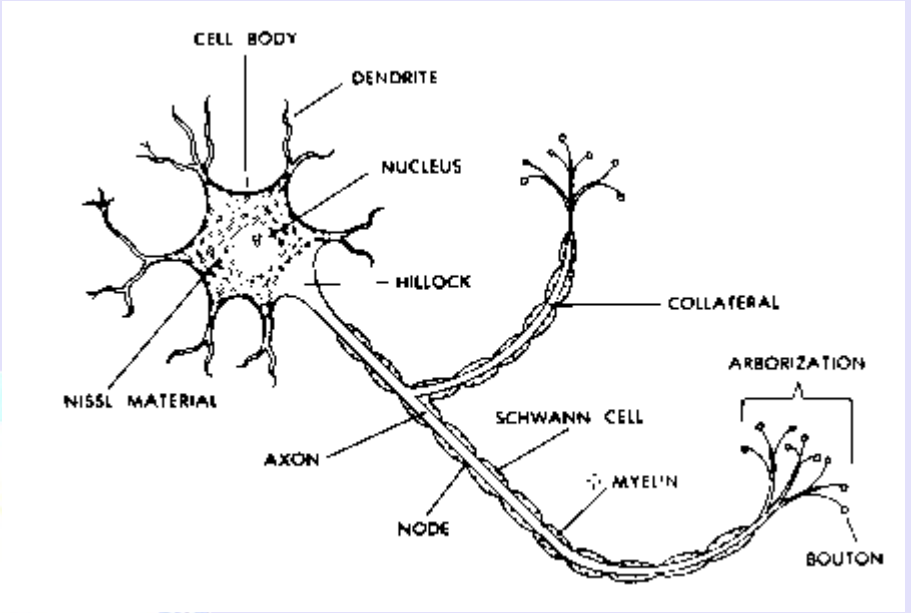
- **Pigments** are **lipofucin** which is yellowish brown pigment common inclusion in nerve cells and **melanin**

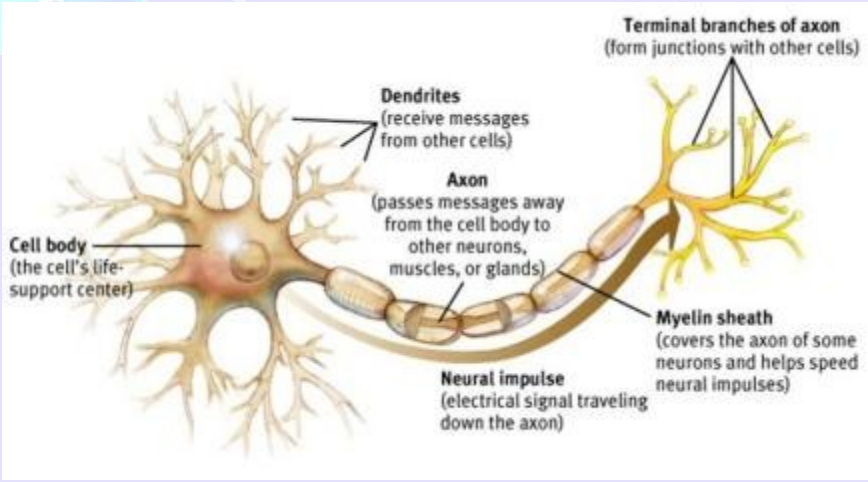
- **Cytoskeleton**: Consists of **microtubules** and **neurofilaments** are randomly scattered in the cytoplasm.
- Intermediate filaments having 10 nm size may not be visible under light microscope



Dendrites

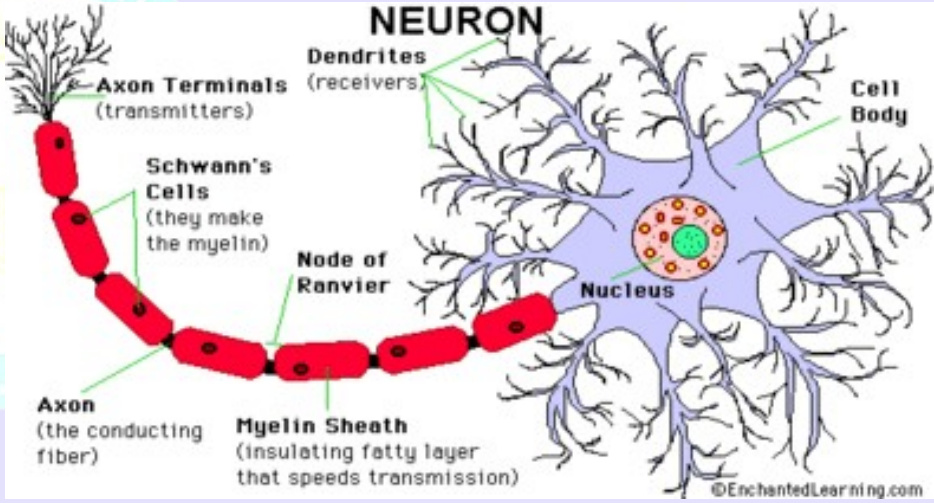
- Afferent processes of neurons
- Have primary secondary and tertiary branches.
- Contains all the components of perikayon except golgi apparatus
- Nissal substance restricted to main stem
- Outer surface shows numerous small spines or knobbed out growths called gemmules these are sites of synaptic contacts.



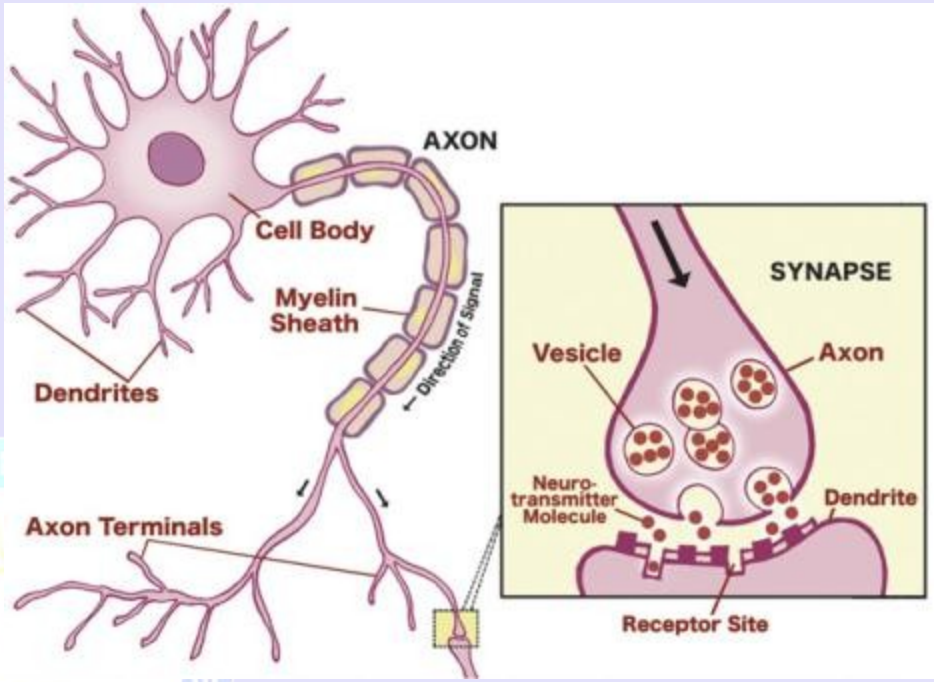


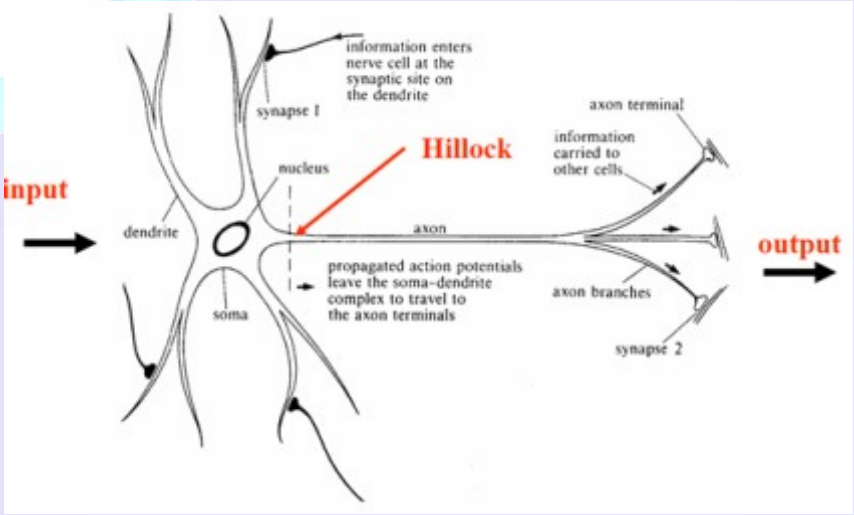
Axons

- Efferent processes, mostly single, start from axon hillock
- Longer, straighter and thinner and have smooth contour and uniform diameter
- Less than few micrometers to millimeters to more than meter in length
- Cytoplasm of axon is called axoplasm and its plasmalemma is called axolemma
- It is surrounded by lipid



- First portion of axon extending from perikaryon to the beginning of myelin sheath is called initial segment
- Axon has uniform diameter and does not branch profusely
- Collaterals are side branches given along its course at right angles
- End by dividing into terminal branches telodendria
- Axon collaterals and telodendria have along their course (bouton en passage) or at their ends

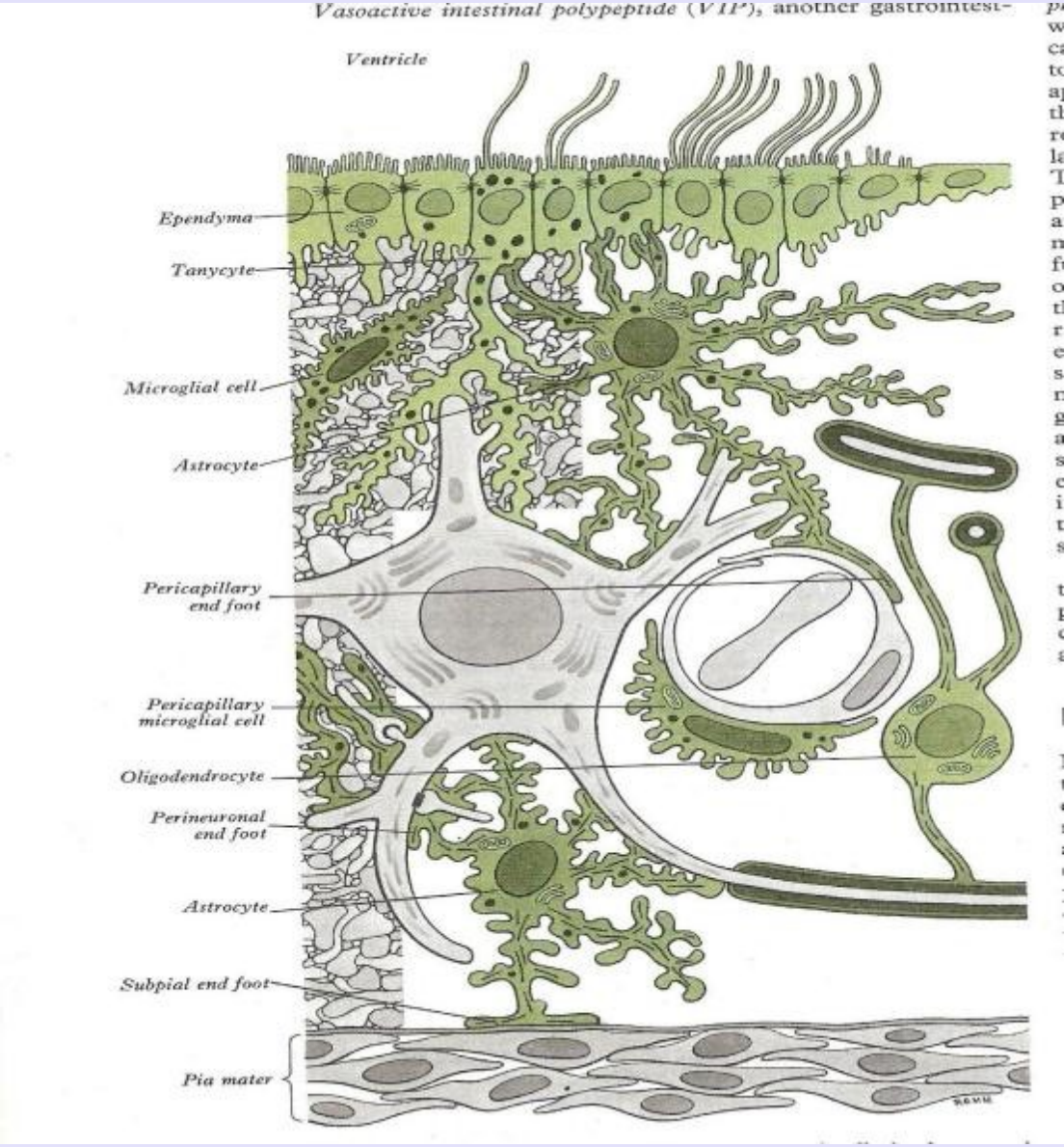




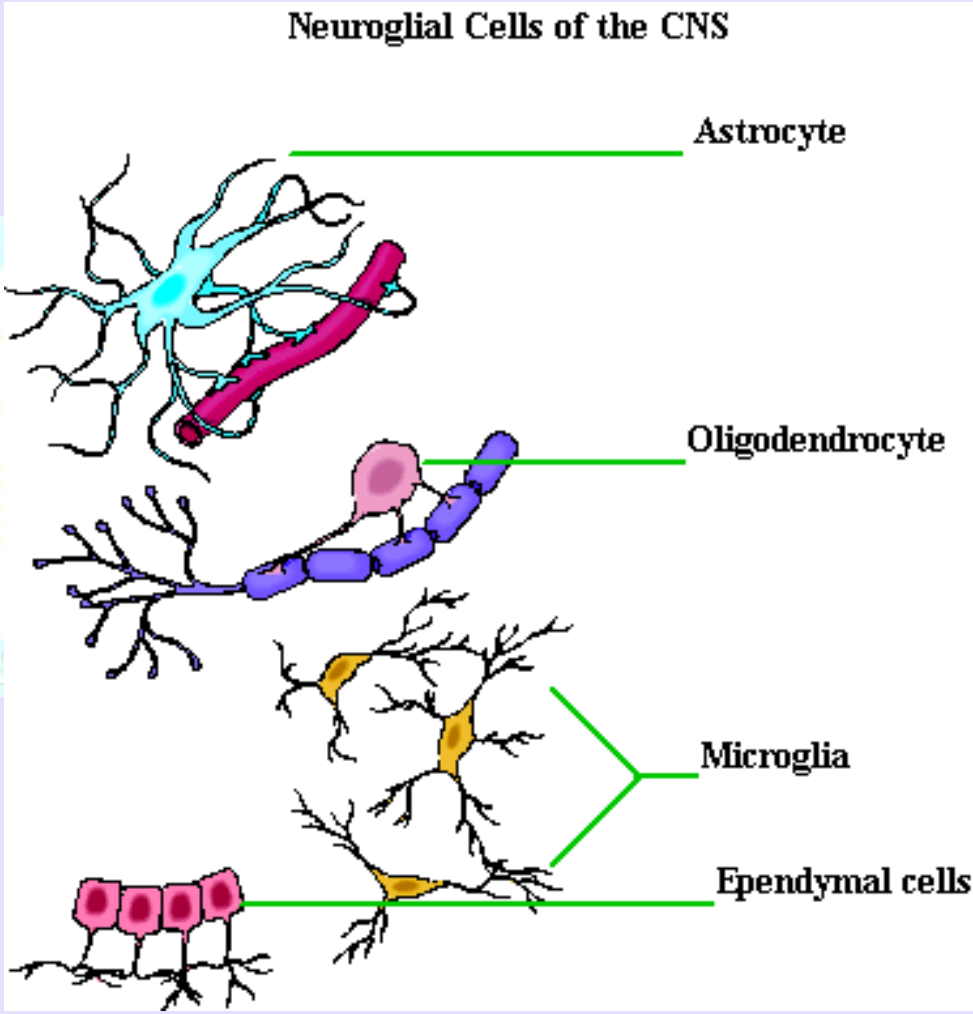
NEUROGLIA

Neuroglial cells a

- Astrocytes
- Microglial cells
- Oligodendrocytes
- Schawan cells
- Ependymal cells

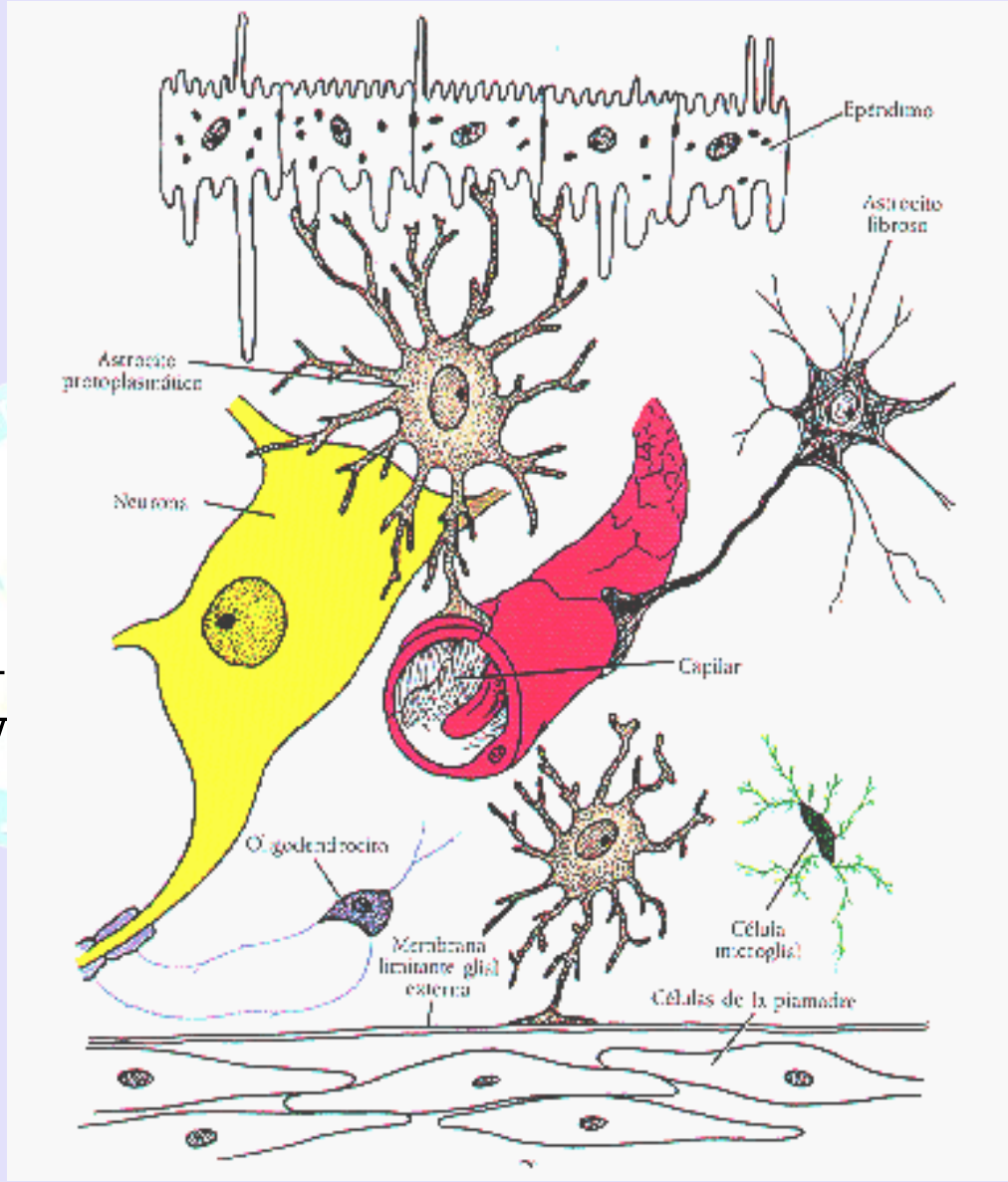


- These are also called supporting cells and are non excitable and are not able to conduct impulses to other cells
- These cells are 10 times more than neurons
- Do not generate action potential and do not synapses with



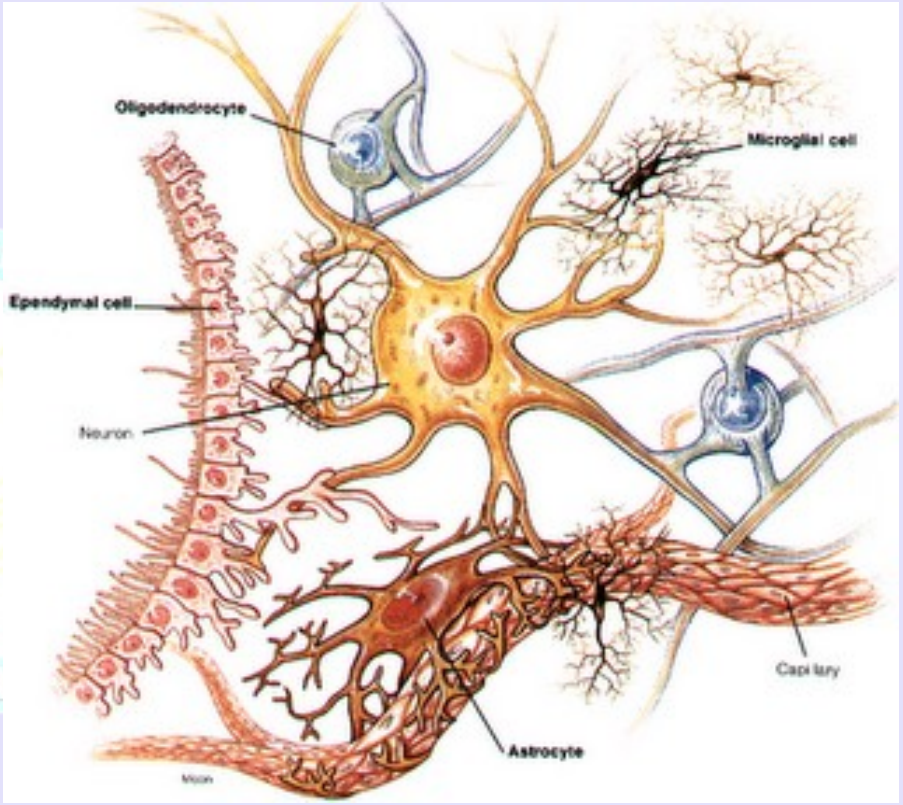
- Their main action is providing supporting frame work for neuron
- Is necessary for the maintenance and viability of neurons
- Play an important role in the metabolic exchange between the neurons and their environment
- Best studied in silver or gold impregnation techniques

- ***Astrocytes:***
- Are star shaped
- It has many branching processes
- Large centrally located spherical nuclei (stain lightly due to its vesicular nature)
- Cytoplasm contains golgi apparatus, lysosome, few ribosomes and small amount of rough endoplasmic reticulum
- Bundles of intermediate filaments extend into the processes



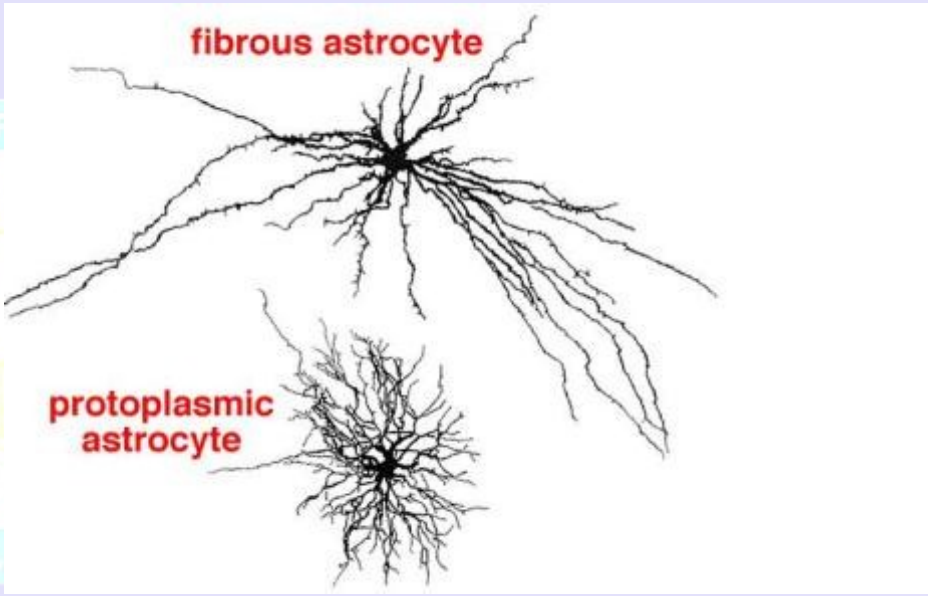
- Some of cytoplasmic processes of astrocytes have expended pedicles at the ends applied to the walls of capillaries called **perivascular feet**
- Processes extend to the surface of the brain and spinal cord forming a layer beneath the pia matter
- Astrocytes may be divided into two groups

1. Protoplasmic astrocytes
2. Fibrous astrocytes



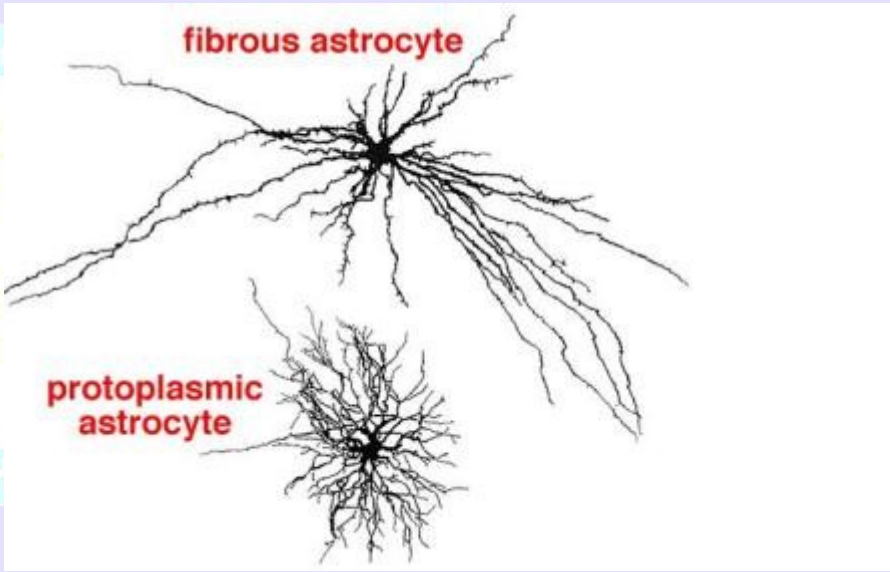
Protoplasmic astrocytes

- Found in the gray matter
- There processes are shorter and thicker but branch extensively.
- Processes envelop the neuron surfaces, synaptic areas and blood vessels.



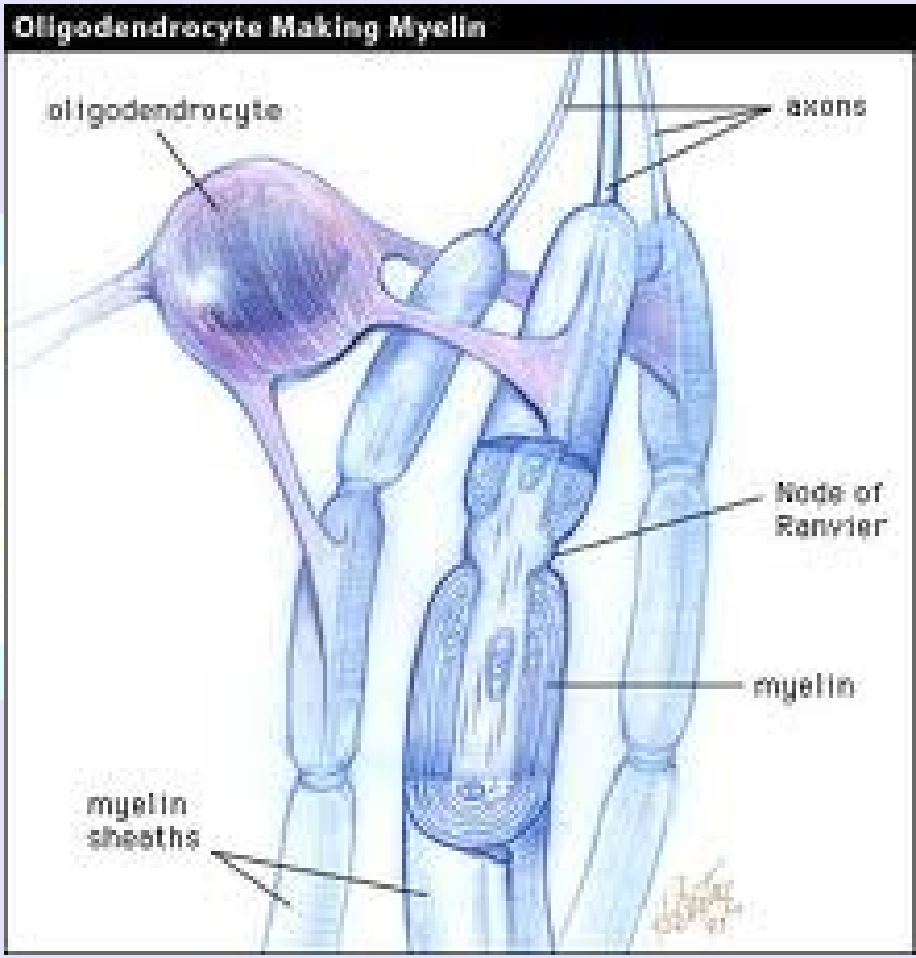
Fibrous astrocytes

- Found in the white matter
- Fewer processes which branch infrequently and are thinner and longer than protoplasmic astrocyte
- Fibrous astrocytes are scaring cells of the CNS
- They fill in the gaps after tissue is lost

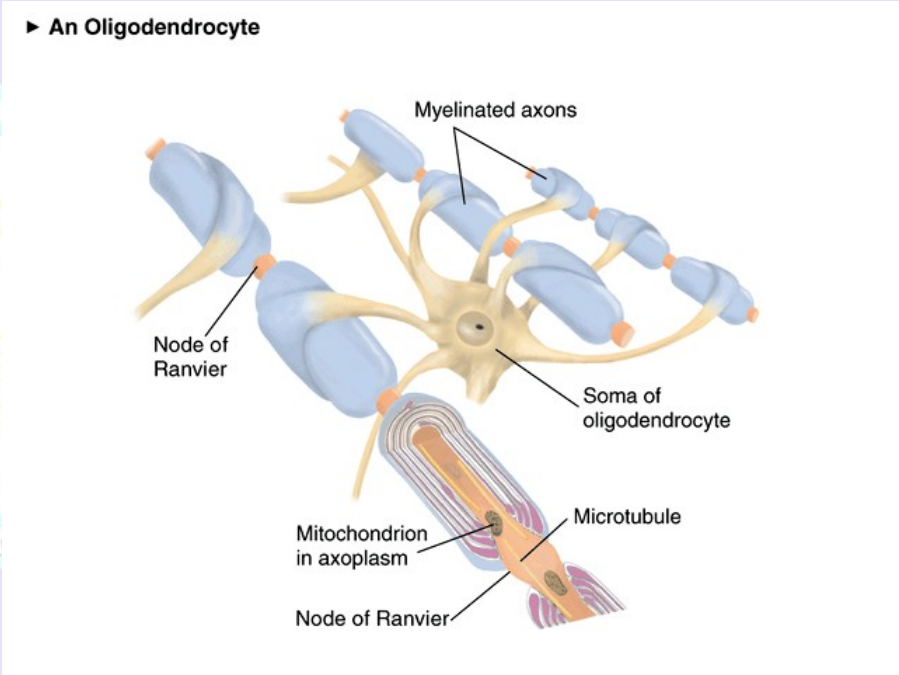


Oligodendrocytes:

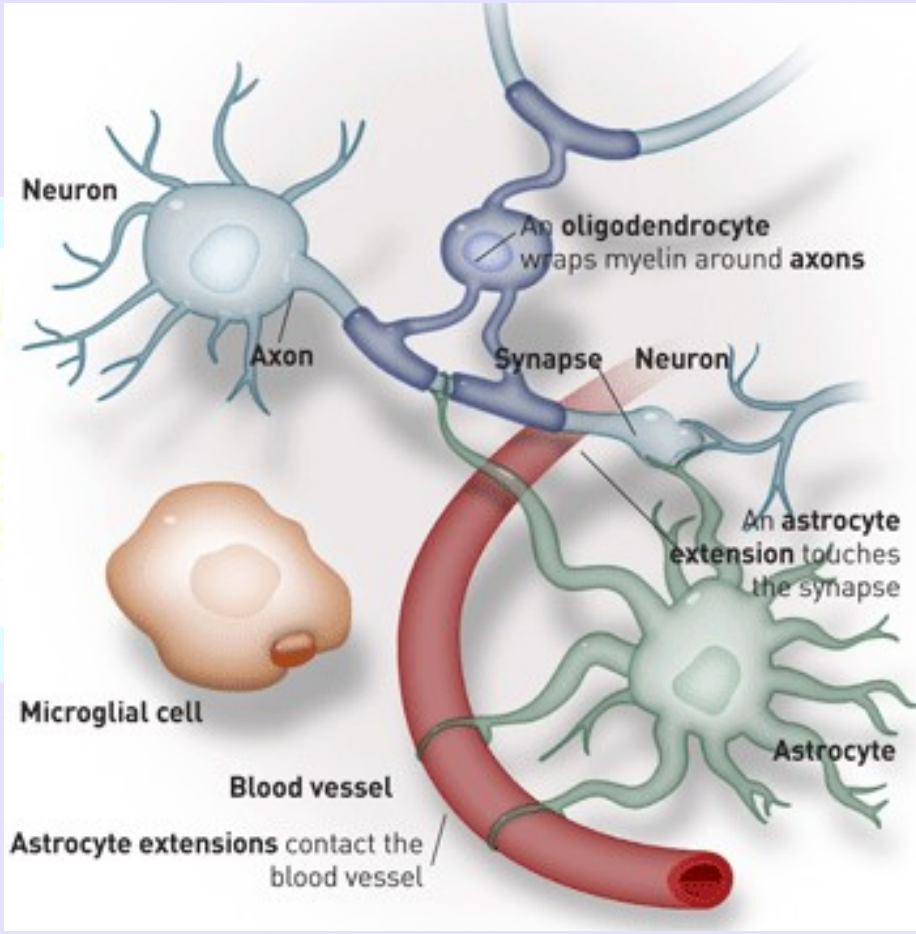
- Smaller cells.
- Smaller, rounder and denser nuclei.
- Processes are fewer and shorter (oligo = scanty, dendron = tree).
- No perivascular feet.



- Cytoplasm contains large number of mitochondria, extensive golgi apparatus, many ribosomes, numerous microtubules
- According to location divided in to three types,
- Interfacicular oligodendrocyes are found in the white mater lying between the nerve fibers.
- Perineural oligodendrocyes are present adjacent to perikaryon of neuron (in the gray mater)
- Perivascular oligodendrocytes found

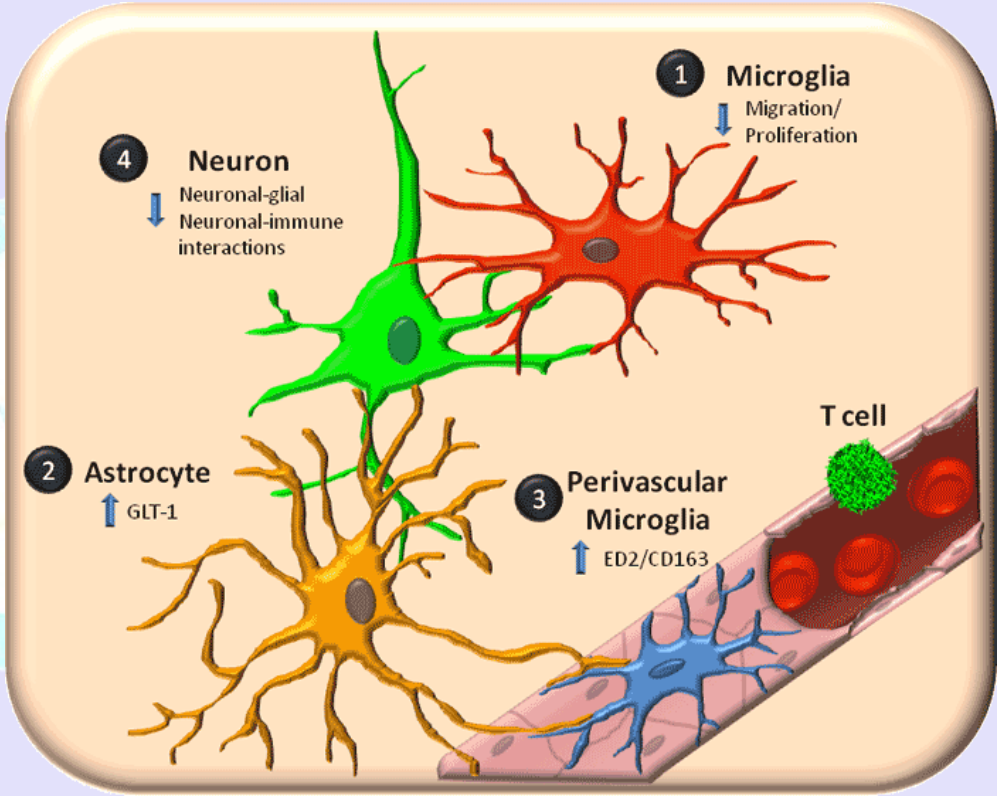


- In perivascular and interfacicular location the oligodendrocytes are responsible for formation of myelin
- Each oligodendrocytes has many processes which form myelin sheaths around several adjacent nerve fibers
- Astrocyte and oligodendrocytes are also called macroglial cells

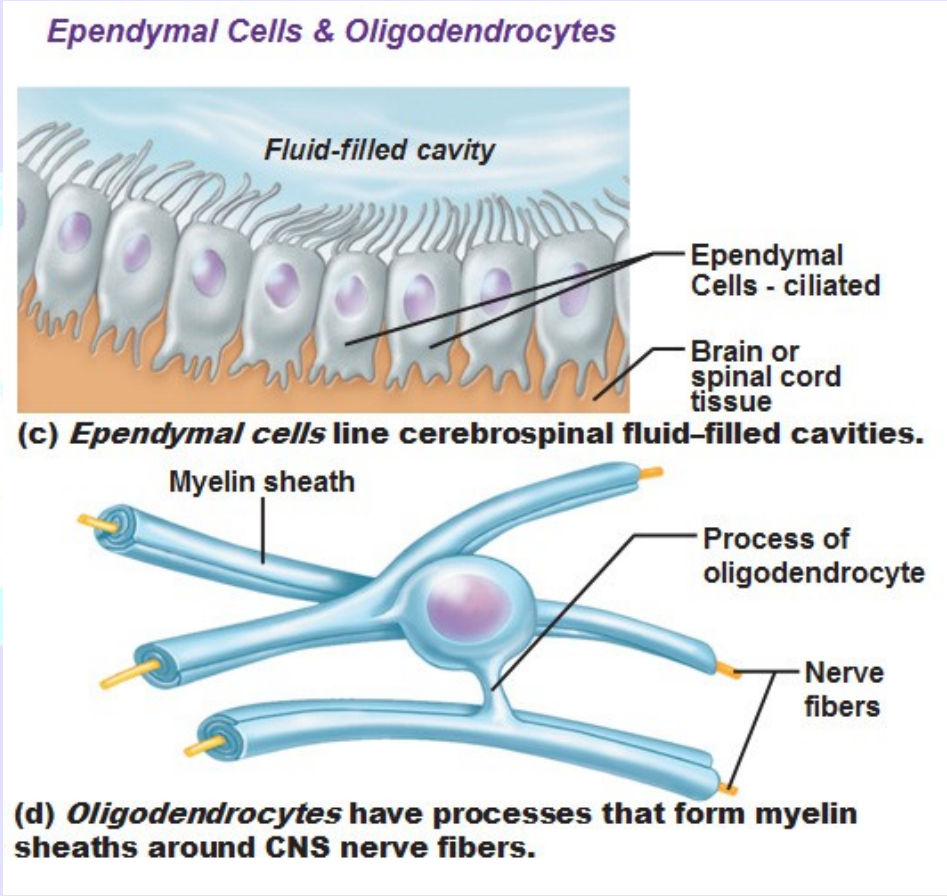


Microglia

- Small, elongated cells
- Rod shaped nuclei which stain deeply due to condensed chromatin
- It has delicate processes which bear small spines.
- Present in both gray and white maters
- Perform phagocytic function
- Mesodermal in origin
- Arise from the pericytes of capillaries or monocytes of blood.



- **Ependymal cells**
- Columnar or cuboidal epithelium which lines the cavities of the brain and spinal cord.
- Are closely packed adjacent cells are held together by desmosomes and junctional complexes.
- Free surfaces of these cells shows numerous microvilli.



THE END